

Troubleshooting Guide: Sulfur Pit Degassing — Ventilation Underperformance and Rising H2S Levels

Category: Safety Systems / Sulfur Handling **Unit:** Liquid Sulfur Pit/Storage, Degassing Ventilation System **Tools:** Ventilation airflow trending (anemometer-verified, not just motor amps), off-gas rate estimate from sulfur production and dissolved H2S content **Fluid Package:** Not applicable — this is an airflow/ventilation capacity balance problem, not a phase-equilibrium calculation **Symptom:** Rising H2S concentration readings at the pit vent and ambient monitors, approaching the site action level

1. Quick Reference

Item	Detail
Reported symptom	H2S concentration at the sulfur pit vent/ambient monitors trending up toward the action level
Initially unclear	Whether this was a change in sulfur production rate/dissolved H2S content, or a ventilation system underperforming despite appearing to run normally
Actual root cause	The degassing ventilation blower's drive belt had worn and was slipping, reducing actual air delivery well below the rate needed to dilute H2S off-gas, even though the blower motor itself was still drawing normal current
Fix	Replaced the worn/slipping belt; verified restored airflow via direct anemometer measurement
Diagnostic signal	Motor amperage looked completely normal (masking the fault), but direct airflow measurement showed a significant shortfall from rated capacity
Prevention	Periodic direct airflow verification via anemometer (not just motor current) added to the PM program; belt condition inspection tied to the airflow trend

2. Symptom

- **H2S concentration readings at the sulfur pit vent and ambient monitors trended upward**, approaching the site's action level — a safety-significant trend requiring prompt investigation.

3. Why This Wasn't Assumed to Be a Production-Rate Issue

Rising H2S readings from a sulfur pit are most directly explained by an increase in dissolved H2S content in the incoming liquid sulfur (e.g., from an upstream Claus/degasging process

change) or an increase in sulfur throughput — both process-side explanations. But the degassing **ventilation system's** job is specifically to dilute and remove H₂S off-gas to keep concentrations below the action level; if that system's actual performance had degraded, the same rising-H₂S symptom would appear even with unchanged sulfur production. Both possibilities needed to be checked.

4. Diagnostic Approach

Step 1 — Review sulfur production rate and upstream dissolved H₂S content

Sulfur throughput and the upstream Claus/degassing process's dissolved H₂S content in the liquid sulfur feed to the pit were reviewed and found essentially unchanged — ruling out a process-side driver for the rising readings.

Step 2 — Check the ventilation blower's operating status

The degassing ventilation blower's control system status and **motor amperage** were checked and found **normal** — the motor was running and drawing its expected current, giving no indication of a problem from the control room.

Step 3 — Directly measure actual airflow delivery

Rather than relying on motor amperage as a proxy for airflow (a proxy that only confirms the motor is running, not that it's actually moving the rated volume of air), a **direct anemometer measurement** was taken at the ventilation discharge.

Finding: Actual airflow was **significantly below the blower's rated capacity**, despite the motor drawing normal current — a clear “commanded vs. delivered” mismatch, conceptually similar to the distillation column thyristor case (Case Study 4), but here on a mechanical belt-drive rather than an electrical heater.

Step 4 — Identify the mechanical cause

With a real airflow shortfall confirmed independent of motor status, the blower's **belt drive** was inspected directly.

Finding: The drive belt was **worn and slipping**, transmitting less power to the fan than the motor was producing — explaining why the motor could draw normal current while the fan itself moved far less air.

Quantitative Basis

- Ventilation blower rated airflow: 4,200 CFM. Direct anemometer measurement during the investigation: **2,650 CFM — a 37% shortfall**, despite motor amperage reading within 2% of its normal baseline value throughout.
- H₂S concentration at the pit vent monitor: baseline 4-6 ppm, rose to **18 ppm** against a site action level of 20 ppm.
- Estimated H₂S off-gas generation rate (from sulfur throughput and dissolved H₂S content, essentially unchanged): consistent with the baseline design case — confirming the ventilation shortfall, not a production change, as the driver.

5. Root Cause

The degassing ventilation blower's drive belt had worn and was slipping, reducing actual air delivery to well below the rate needed to adequately dilute H₂S off-gas from the sulfur

pit — even though the blower motor itself continued to run and draw normal current, giving no indication of the underlying mechanical shortfall from standard control-room monitoring alone.

6. Corrective Action

1. **Replaced the worn/slipping drive belt.**
2. **Verified restored airflow via direct anemometer measurement** rather than relying on motor amperage alone.

7. Verification

- Post-repair anemometer measurement confirmed airflow at **4,150 CFM — 99% of the 4,200 CFM rated capacity.**
- H₂S concentration at the pit vent monitor returned to **5 ppm**, within the normal 4-6 ppm baseline range, well below the 20 ppm action level.
- Held stable over the following **30 days** of continued monitoring.

8. Prevention / Long-Term Fix

- Added **periodic direct airflow verification via anemometer** to the PM program, specifically because motor amperage alone had proven insufficient to detect this failure mode.
 - **Belt condition inspection is now tied to the airflow trend** (triggered by any deviation, not just a fixed calendar interval), since belt wear can progress without producing any electrical-side symptom.
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9. General Troubleshooting Checklist (Reusable)

- ☐ Review process-side factors (production rate, upstream dissolved H₂S content) first to rule in/out a genuine process change
- ☐ Check ventilation/blower control status and motor amperage, but recognize that normal motor current does NOT confirm normal air delivery
- ☐ Take a direct airflow measurement (anemometer) at the ventilation discharge to verify actual delivered performance against rated capacity
- ☐ If a shortfall is confirmed despite normal motor amperage, inspect the mechanical power transmission path (belts, couplings, fan blade condition) rather than the motor/electrical side
- ☐ Repair the mechanical fault and re-verify with a direct airflow measurement, not just a return-to-normal motor amperage reading
- ☐ Confirm H₂S/ambient monitoring readings return to baseline and hold stable over an extended monitoring period
- ☐ Add direct airflow verification (not just motor current) to the standing PM program for any ventilation system protecting against a toxic gas exposure risk

10. Key Takeaway

Motor amperage tells you the motor is running — it does not tell you the fan is actually moving its rated volume of air. A slipping belt, damaged fan blade, or other mechanical power-transmission fault can leave a ventilation system silently underperforming for a safety-critical duty while every electrical indicator looks completely

normal. For any ventilation system protecting against a toxic gas hazard, periodic direct airflow measurement is not optional — it's the only way to actually confirm delivered performance, not just commanded performance.

Related Concepts / Tags

sulfur-pit degassing H₂S ventilation-system blower-belt airflow-verification anemometer safety-critical-ventilation toxic-gas-exposure

This guide is derived from a representative field troubleshooting scenario. Values and thresholds shown are illustrative and should be validated against your own unit's design basis before applying.